AI-DRIVEN COMPLIANCE ASSESSMENT

INTRODUCTION

Build a dynamic, data-driven compliance system that leverages machine learning to predict and flag potential issues before the shipment is dispatched. The model adapts by learning from historical compliance issues and continuously improves its accuracy.

PROPOSED SOLUTON

- Historical Data Collection:Aggregate past shipment data, including flagged compliance issues.
- Data Preprocessing:Clean and normalize data for consistent ML model training.
- Model Training:Use classification algorithms (e.g., random forest, gradient boosting) to predict compliance issues.
- Real-Time Scoring:Ingest new shipment data, score it for risk, and flag potential issues.
- User Interaction:Provide a ranked list of shipments based on risk score via a simple UI.

TECH STACK

- Backend: Python with frameworks like TensorFlow or PyTorch for model building.
- Database: SQL/NoSQL hybrid for historical and real-time data.
- Frontend: React or Angular for the dashboard.
- DevOps: CI/CD pipelines integrated with cloud services (AWS SageMaker for ML deployment).

VIABILITY AND FEASIBILITY

- Viability: The predictive model can significantly reduce compliance issues by preemptively flagging risky shipments.
- Feasibility: With access to sufficient historical data, training and deploying the ML model is straightforward. Regularupdates and feedback mechanisms ensure that the system remains accurate and effective in real time.

Thank you